an end destination, or it could be getting to the next step on a set of directions. For this, you can give the user the next instruction.

To accomplish this, `LocationManager` offers `addProximityAlert`(). This registers a PendingIntent, which will be fired off when the device gets within a certain distance of a certain location. `TheaddProximityAlert`() method takes the following as parameters:

- The latitude and longitude of the position of interest
- A radius, specifying how close you should be to that position for the Intent to be raised
- A duration for the registration, in milliseconds (after this period, the registration automatically lapses); a value of -1 means the registration lasts until you manually remove it via `removeProximityAlert`()
- The PendingIntent to be raised when the device is within the target zone expressed by the position and radius

Note that there's no guarantee that you'll actually receive an Intent. There may be in an interruption in location services, or the device may not be in the target zone during the period of time the proximity alert is active. For example, if the position is off by a bit, and the radius is a little too tight, the device might only skirt the edge of the target zone, or it may go by the target zone so quickly that the device's location isn't sampled during that time.

It's up to you to arrange for an activity or intent receiver to respond to the Intent you register with the proximity alert. What you do when the Intent arrives is up to you. For example, you might set up a notification (e.g., vibrate the device), log the information to a content provider, or post a message to a web site.

Note that you'll receive the Intent whenever the position is sampled and you're within the target zone, not just upon entering the zone. Hence, you'll get the Intent several times—perhaps quite a few times, depending on the size of the target zone and the speed of the device's movement.

4.5. Testing

The Android emulator doesn't have the ability to get a fix from GPS, triangulate your position from cell towers, or identify your location through nearby Wi-Fi signals. So, if you want to simulate a moving device, you'll need to have some means of providing mock location data to the emulator. One likely option for supplying mock location data is the Dalvik Debug Monitor Service (DDMS). This is an external program, separate from the emulator, which can feed the emulator single location points or full routes to traverse, in a few different formats.